

AI-Powered Logistics Decision Engine – Project Journey

Project Goal

I wanted to build a machine learning system that could predict shipment delays and help logistics teams make better delivery decisions.

Instead of creating a simple classification model, my goal was to understand the data, investigate business patterns, and build a recommendation system that could suggest the best delivery mode.

Step 1: Understanding the Dataset

The dataset contained 25,000 shipment records with information such as:

- Delivery Partner
- Package Type
- Vehicle Type
- Delivery Mode
- Region
- Weather Condition
- Distance
- Package Weight
- Expected Delivery Time
- Delivery Status
- Delivery Cost

Before training any model, I explored the dataset structure, data types, and class distribution.

Step 2: Exploratory Data Analysis (EDA)

I analyzed:

- Delay rate across delivery partners
- Delay rate across vehicle types
- Delay rate across delivery modes
- Weather impact on delays
- Distance and package weight distributions
- Target class balance

During this process, I discovered that approximately 26.7% of shipments were delayed.

I also noticed some unusual patterns in the data, which led me to investigate further.

Step 3: Feature Engineering

To improve model performance, I created additional business-oriented features.

Heavy Package Flag

Indicates whether a package weighs more than 10 kg.

Long Distance Flag

Indicates whether a shipment travels more than 100 km.

Load Effort

A custom feature defined as:

Load Effort = Distance × Package Weight

This feature represents the overall transportation effort required for a shipment.

Step 4: Data Preprocessing

I prepared the dataset for machine learning by:

- Encoding categorical features using One-Hot Encoding
- Scaling numerical features using StandardScaler
- Creating a preprocessing pipeline using ColumnTransformer

This ensured that all models received properly transformed input data.

Step 5: Model Development

I trained and compared three machine learning models:

1. Logistic Regression
2. Random Forest
3. XGBoost

Using a train-test split with stratified sampling, I evaluated each model using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Results showed that all three models achieved approximately 89% accuracy and strong F1 scores.

Step 6: Investigating Suspicious Patterns

While analyzing the data, I discovered several unexpected behaviors.

Examples:

- Some shipments appeared to have 0-hour delivery times.
- Express delivery showed higher delay rates than standard delivery.
- Expected delivery time strongly influenced delay outcomes.

Instead of ignoring these findings, I investigated them through:

- Crosstab analysis
- Distribution analysis
- Feature importance analysis
- Controlled experiments

This helped me understand that the dataset likely contains synthetic or generated business rules.

Step 7: Feature Importance Analysis

Using Random Forest feature importance, I identified the most influential variables.

Top features included:

- Expected Delivery Time
- Delivery Mode
- Distance
- Load Effort
- Package Weight

This provided valuable insight into the model's decision-making process.

Step 8: Business Analysis

Beyond model accuracy, I analyzed operational performance.

I compared:

Delivery Partners

- Delay Rate
- Average Delay
- Average Cost

Vehicle Types

- Delay Rate

- Average Delay
- Average Cost

Delivery Modes

- Delay Rate
- Average Delay
- Average Cost

This transformed the project from a pure machine learning exercise into a business analytics project.

Step 9: Building a Recommendation Engine

Instead of stopping at delay prediction, I built a recommendation system.

For a given shipment, the system evaluates:

- Standard Delivery
- Two-Day Delivery
- Same-Day Delivery
- Express Delivery

For each option, the model predicts:

- Delay Probability
- Risk Level

The system then recommends the delivery mode with the lowest operational risk.

Step 10: Decision Optimization

To make recommendations more realistic, I introduced:

- Cost estimation
- Speed scoring
- Decision scoring

This allowed the system to balance:

- Delivery Risk
- Delivery Cost
- Delivery Speed

similar to real-world logistics decision-making.

Challenges Faced

1. Understanding unexpected patterns in the dataset.

2. Detecting potential data leakage.
 3. Investigating unusual delivery-time values.
 4. Balancing business logic with machine learning predictions.
 5. Converting a classification model into a practical decision-support system.
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Key Learnings

Through this project I learned:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Machine Learning Pipelines
- Model Evaluation
- Feature Importance Analysis
- Business Analytics
- Recommendation System Design
- Risk Scoring
- Decision Support Systems

Most importantly, I learned that building a machine learning model is only part of the process. Understanding the data, validating assumptions, and solving a real business problem are equally important.